THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET D

Chapters 8 & 9: **Introduction to Trigonometry, Some Applications of Trigonometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures for questions on heights and distances. Take $\sqrt{3} = 1.732$ where needed.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\sec 60^\circ$ equals: 1

(a) 2 **(b)** $\frac{1}{2}$ **(c)** $\sqrt{3}$ **(d)** $\frac{2}{\sqrt{3}}$

2. If $\tan A = \frac{3}{4}$, then $\cos A$ is: 1

(a) $\frac{4}{5}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{5}{4}$ **(d)** $\frac{4}{3}$

3. $(\sin 30^\circ + \cos 30^\circ) - (\sin 60^\circ + \cos 60^\circ)$ equals: 1

(a) −1 **(b)** 0 **(c)** $\sqrt{3}$ **(d)** 1

4. A student states: "$\sin^2\theta$ means $\sin(\theta^2)$." The correct response is: 1

(a) Incorrect — it means $(\sin\theta)^2$ **(b)** Correct

(c) Both mean the same **(d)** Cannot be determined

5. For $\sec^2 45^\circ - \tan^2 45^\circ$, Student A says 1; Student B says 0. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. A 20 m ladder against a wall makes 60° with the ground. Its foot is at a distance of: 1

(a) 5 m **(b)** $10\sqrt{3}$ m **(c)** 20 m **(d)** 10 m

7. When the sun's elevation is 30°, a 5 m pole casts a shadow of: 1

(a) 5 m **(b)** $\frac{5}{\sqrt{3}}$ m **(c)** 10 m **(d)** $5\sqrt{3}$ m

8. A kite string 80 m long makes 30° with the ground. The height of the kite is: 1

(a) 80 m **(b)** 20 m **(c)** $40\sqrt{3}$ m **(d)** 40 m

9. A student states: "The line of sight is the line from the observer's eye to the object." The correct response is: 1

(a) Incorrect **(b)** It is the horizontal line

(c) Correct **(d)** Cannot be determined

10. From the top of a 30 m building, the angle of depression of a point is 60°. Student A says its distance is $10\sqrt{3}$ m; Student B says $30\sqrt{3}$ m. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $7\sin^2\theta + 3\cos^2\theta = 4$ (θ acute), find $\tan\theta$. 2

12. Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \csc 30^\circ}$. 2

13. Find the length of the shadow of a 6 m pole when the sun's elevation is 60°. 2

14. An aeroplane at a height of 1000 m is seen at an angle of elevation of 30° from A. Find its horizontal distance from A. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sin^4 A - \cos^4 A = 1 - 2\cos^2 A$. 3

16. From a window 12 m above the ground, the angles of elevation and depression of the top and the foot of a building across the street are 60° and 45°. Find the height of the building. 3

17. Prove that $\frac{\cos A}{1+\sin A} = \sec A - \tan A$. 3

OR

If $\csc\theta - \sin\theta = l$ and $\sec\theta - \cos\theta = m$, prove that $l^2m^2(l^2 + m^2 + 3) = 1$. 3

18. Two men on the same side of a 75 m tower observe its top at 30° and 60°. Find the distance between them. 3

OR

An observer 1.5 m tall is 28.5 m from a 30 m tower. Find the angle of elevation of the top from his eyes. How far should he be for the angle to be 60°? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that $\frac{1}{\sec A - \tan A} - \frac{1}{\cos A} = \frac{1}{\cos A} - \frac{1}{\sec A + \tan A}$. 5

OR

Prove that $(1+\tan A)^2 + (1-\tan A)^2 = 2\sec^2 A$. Find its value for $A = 60^\circ$. 5

20. **Case study:** A ropeway runs from the foot of a hill to its top along a straight cable 2 km long, making 30° with the horizontal. 5

(i) Find the height of the hill.

(ii) Find the horizontal distance covered.

(iii) If the cable car moves at 4 km/h, how long does the ride take?

OR

A bird is sitting on top of a 20 m tree. From a point on the ground its angle of elevation is 45°. The bird flies horizontally away and after 2 seconds the angle becomes 30°. Find the speed of the bird. 5

END OF QUESTION PAPER — Total: 40 Marks